CH32V205 Evaluation Board Reference

Version: V1.1

https://wch-ic.com

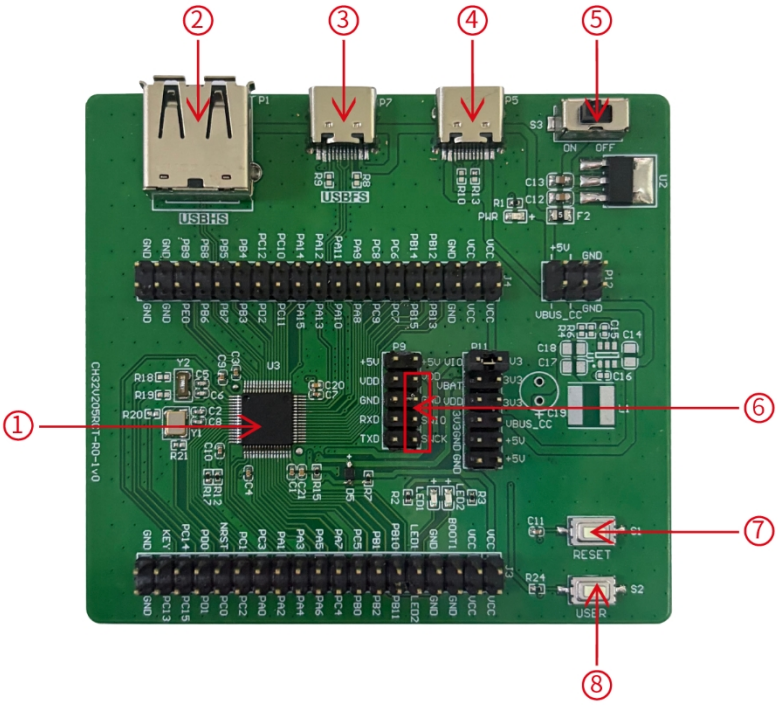
1. Overview

This evaluation board is applied to the development of the CH32V205 chip. The IDE uses the MounRiver compiler, with the option of using the on-board or independent WCH-Link for emulation and download, and provides reference examples and demonstrations of chip resource-related applications.

2. Evaluation board hardware

Please refer to the CH32V205SCH.pdf document for the schematic of the evaluation board.

CH32V205 Evaluation Board



Descriptions

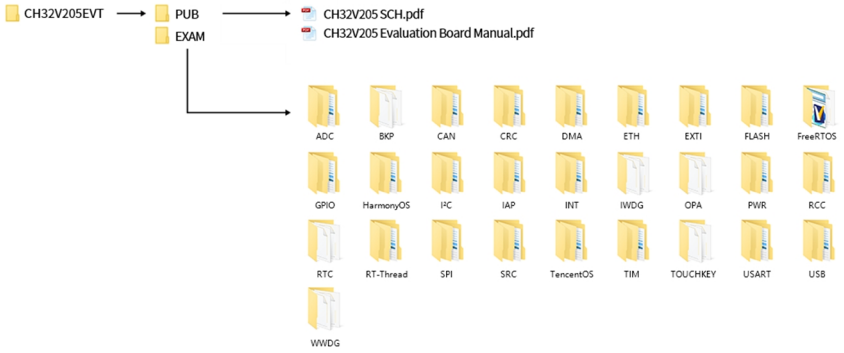
1. Main control MCU	2. USBHS interface	3. USBFS interface	4. USBPD interface
5. Power switch S3	6. Debug interface	7. Reset button S1	8. User button S2

Motherboard - CH32V205-R0

1. MCU: CH32V205RCT
2. USBHS interface P1: USBHS interface PB6, PB7 of the MCU
3. USBFS interface P7: USBFS interface PA11, PA12 of the MCU
4. USBPD interface P5: USBPD interface PA0, PA1 of the MCU
5. Power switch S3: Used to disconnect or connect external 5V power supply or USB power supply
6. Debug interface: download / debug MCU
7. Reset button S1: for reset MCU
8. User button S2: Connect the I/O port of the main MCU for key control

3. Software Development

3.1 EVT package directory structure



Description:
PUB folder: provides evaluation board manuals, evaluation board schematics.
EXAM folder: Provides software development drivers and corresponding examples for the CH32V205 controller, grouped by peripheral. Each type of peripheral folder contains one or more functional application routines folders.

3.2 IDE use-MounRiver

Download MounRiver_Studio, double click to install it, and you can use it after installation. (MounRiver_Studio instructions are available at the path: MounRiver\MounRiver_Studio\ MounRiver_Help.pdf and MounRiver_ToolbarHelp.pdf)

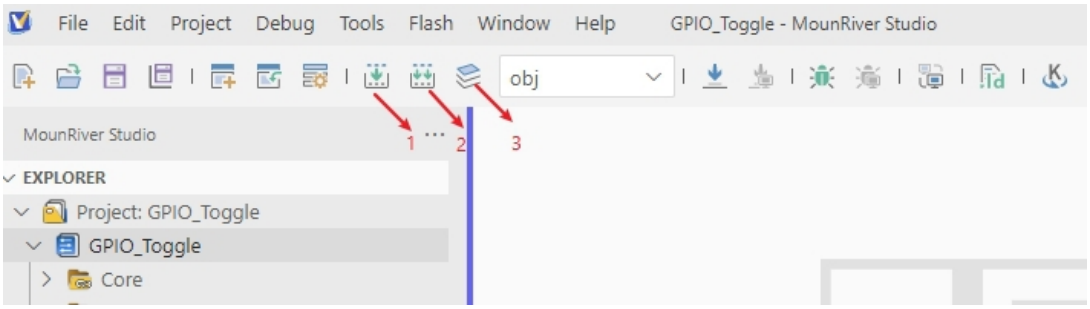
3.2.1 Open project

➢ Open project:

1) Double-click project file directly with the suffix name .wvproj under the corresponding project path.
2) Click File in MounRiver IDE, click Load Project, select the .project file under the corresponding path, and click Confirm to apply it.

3.2.2 Compilation

MounRiver contains three compilation options, as shown in the following figure.



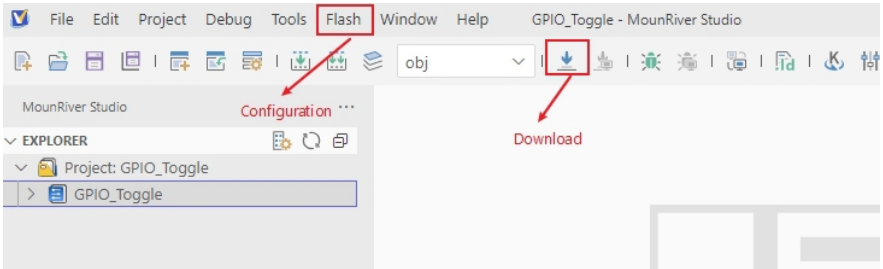
Compile option 1 is Incremental Build, which compiles the modified parts of the selected project.
Compile option 2 is ReBuild, which performs a global compilation of the selected project.
Compile option 3 is All Build, which performs global compilation for all projects.

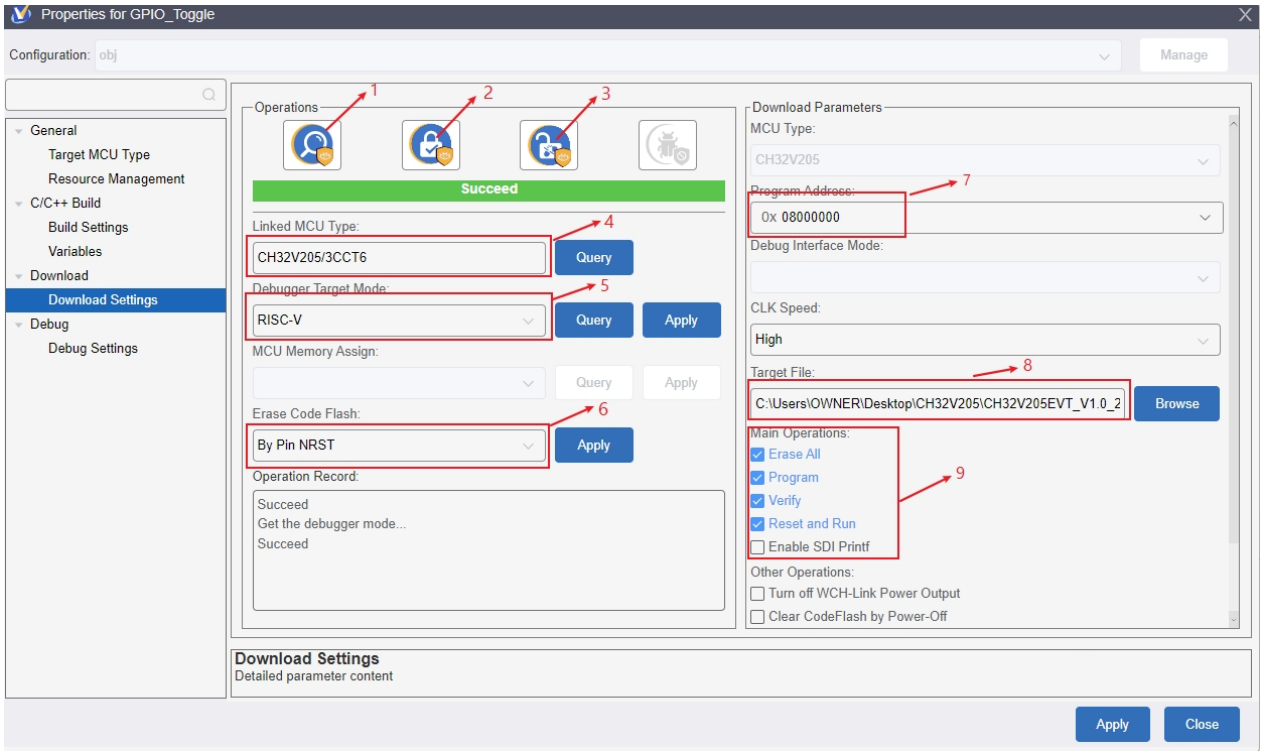
3.2.3 Download/Simulation

➢ Download

1) Debugger download

Connect to the hardware via WCH-Link (see WCH-Link instructions for details, path: MounRiver\MounRiver_Studio\ WCH-Link instructions.pdf), click the Download button on the IDE, and select Download in the pop-up interface, as shown in the figure below.



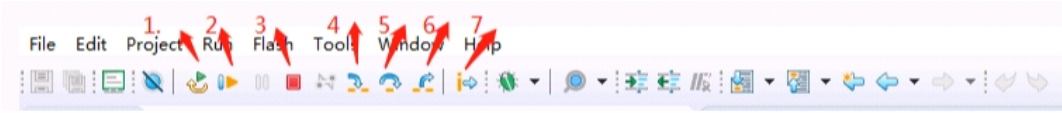


1. Query the chip read protection status;
2. Set the chip read protection, re-power on the configuration to take effect;
3. Release the chip read protection, re-power on the configuration to take effect;
4. Query the display of the current chip type
5. Configure the Link mode
6. power-off to erase the entire user area flash
7. Set the download address
8. Select target file
9. Select download configuration

➢ Simulation

1) Toolbar description

Click Debug button in the menu bar to enter the download, see the image below, the download toolbar.



Detailed functions are as follows.

(1) Restart: if restart, the program will returns to the beginning.

(2) Continue: Click to continue debugging.

(3) Terminate: Click to exit debugging.

(4) Single-step jump-in: Each time you tap a key, the program runs one step and encounters a function to enter and execute.

(5) Single-step skip: jump out of the function and prepare the next statement.

(6) Single-step return: return the function you jumped into

Instruction set single-step mode: click to enter instruction set debugging (need to use with 4, 5 and 6 functions).

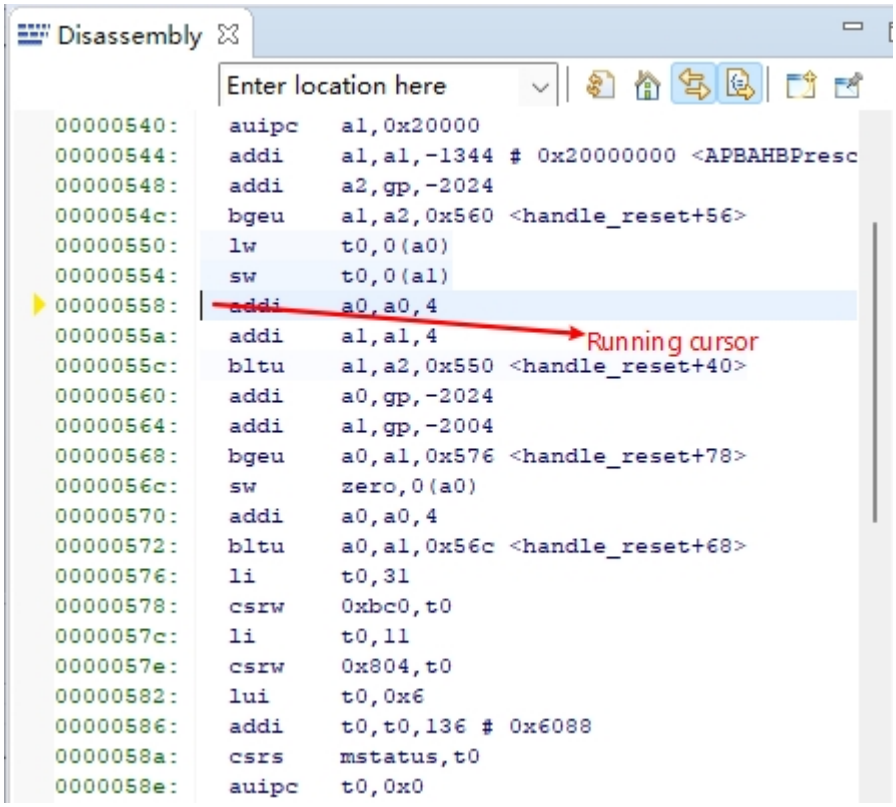
2) Set breakpoints

Double-click on the left side of the code to set a breakpoint, double click again to cancel the breakpoint, set the breakpoint as shown in the following figure;

3) Interface display

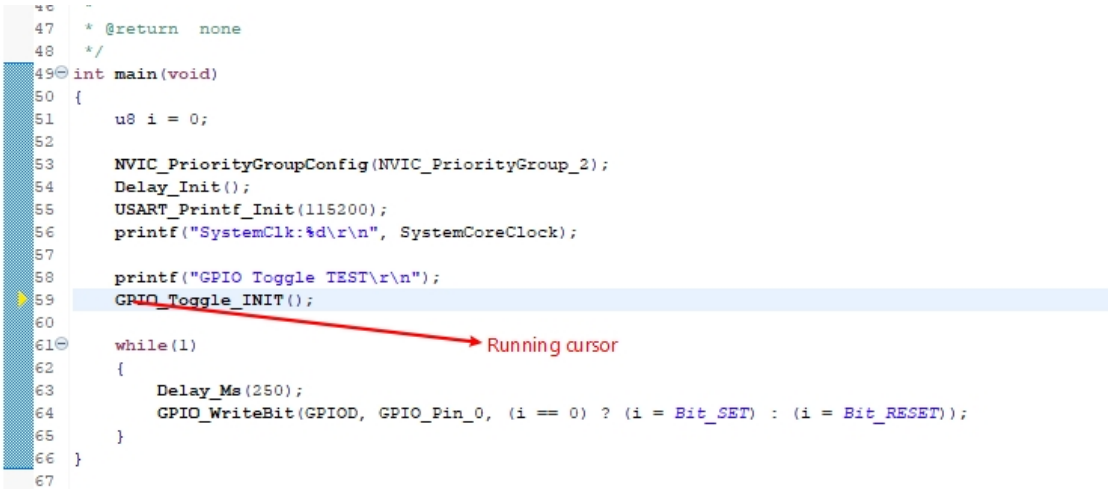
(1) Instruction set interface

Click on the instruction set single-step debugging can enter the instruction debugging, to single-step jump in for example, click once to run once, the running cursor will move to view the program running, the instruction set interface is shown as follows.



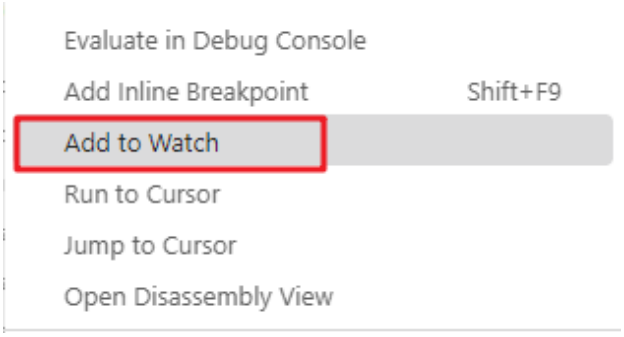
(2) Program running interface

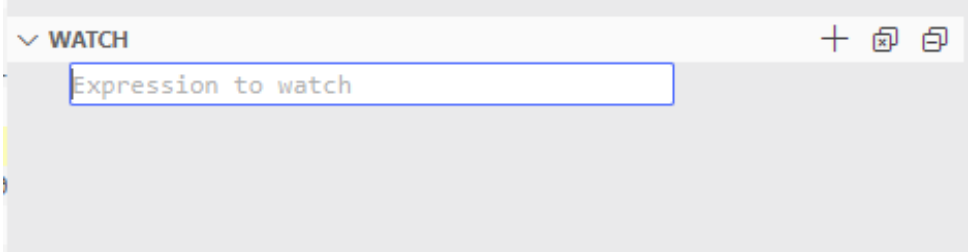
It can be used with instruction set single-step debugging, still take single-step jumping in as an example, click once to run once, the running cursor will move to view the program running, the program running interface is shown as follows.



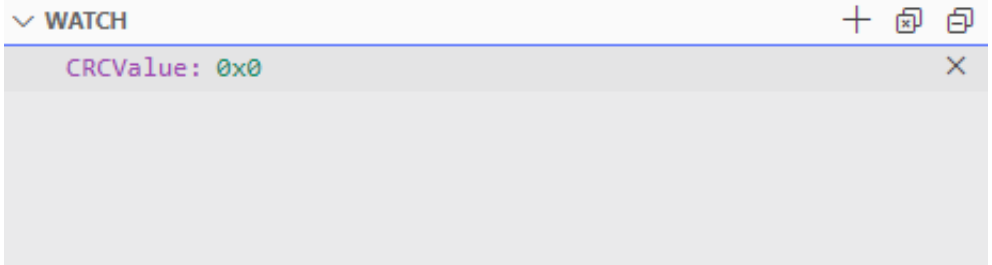
4) Variables

Hover over the variable in the source code to display the details, or select the variable and right-click add watch expression



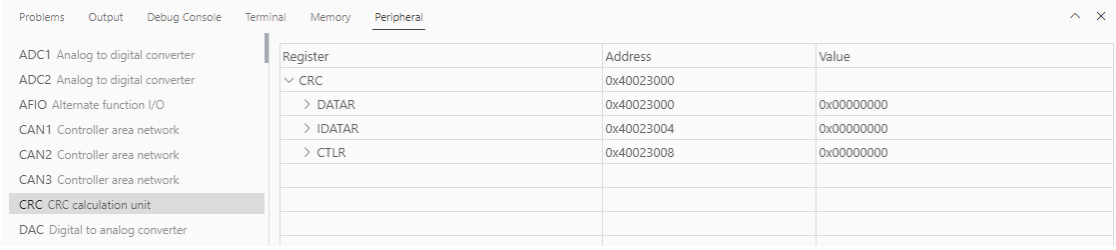


Fill in the variable name, or just click OK to add the variable you just selected to the pop-up.



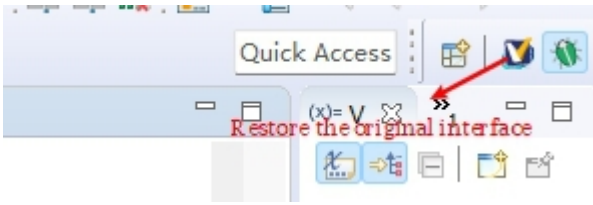
5) Peripheral registers

In the lower left corner of IDE interface Peripherals interface shows a list of peripherals, tick the peripherals will display its specific register name, address, value in the Memory window.



Note:

(1) When debugging, click the icon in the upper right corner to enter the original interface.



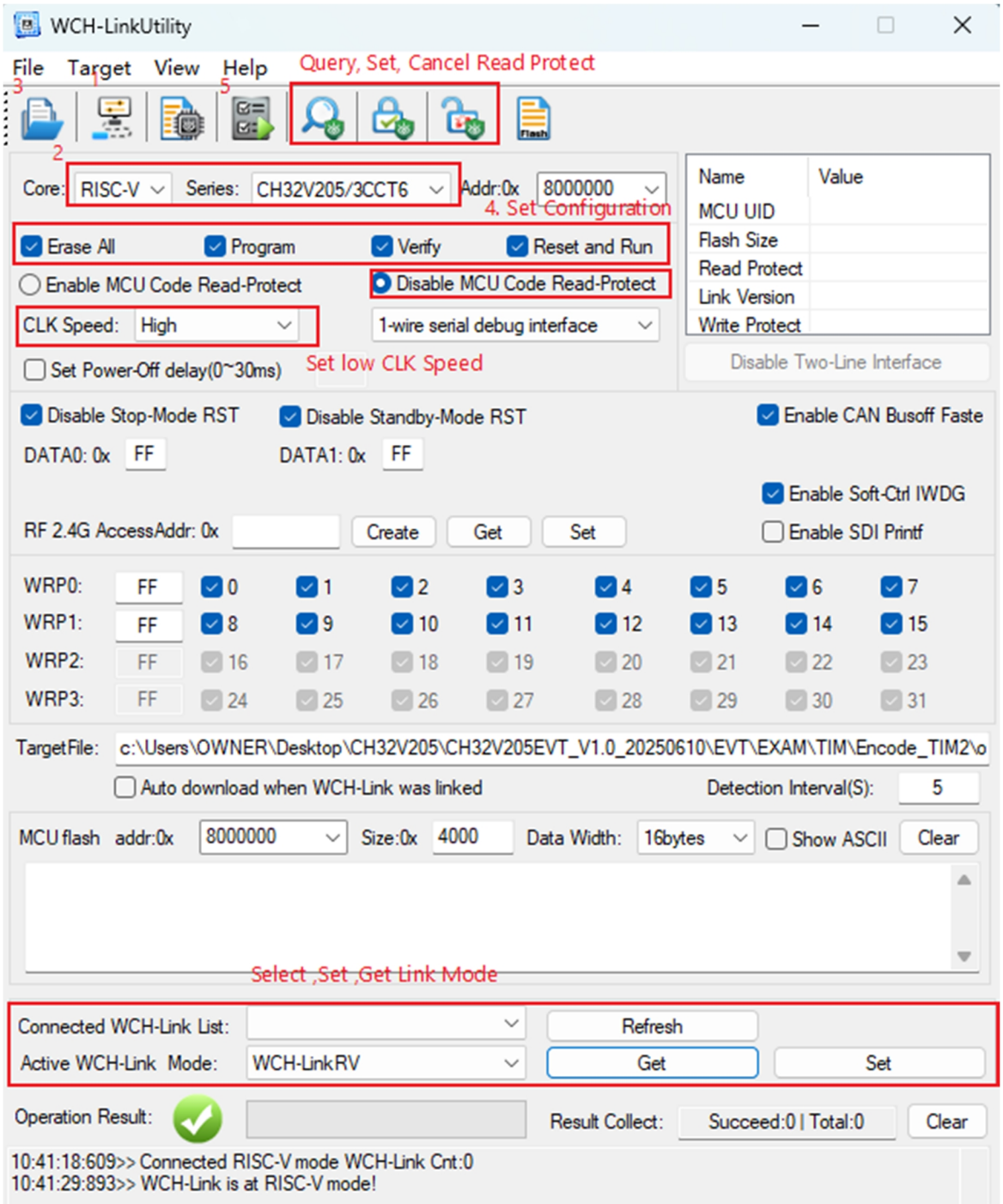
(2) For documentation to access the compiler, click F1 to access the help documentation for detailed instructions.

4. WCH-LinkUtility.exe Download

The download process for the chip using the WCH-LinkUtility tool is:

1) Connect WCH-Link
2) Select chip information

3) Add firmware
4) If the chip is read protected, you need to release the chip read protection.
5) Execute



5. WCHISPTool.exe Download

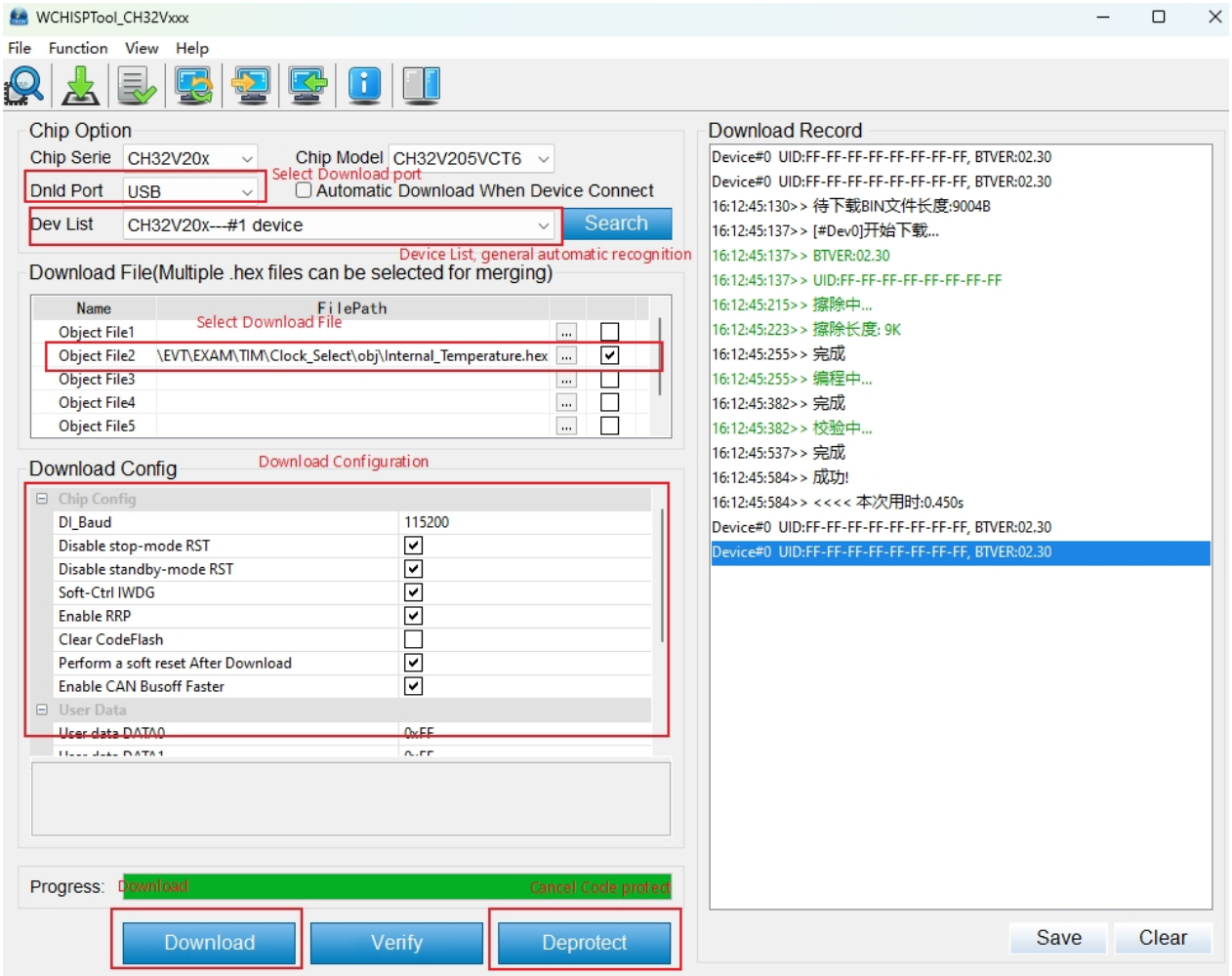
The WCHISPTool tool is used to download the chip, supporting both USB and serial port. The USB pins are PA11 (DM), PA12 (DP), and the serial port pins are PA2 (TX), PA3 (RX). The download process is:

(1) BOOT0 is connected to VCC, BOOT1 is connected to ground, and the PC is connected via serial port or USB.

(2) Open the WCHISPTool tool, select the appropriate download method, choose to download the firmware, check the chip configuration and click download.

(3) Ground BOOT0, reapply power and run APP program.

The WCHISPTool tool interface is shown in the following figure.



1. Select the Serial/USB port download mode
2. Identify the device, usually automatically, if it fails to identify, you need to select manually
3. Select the firmware, .hex or .bin file
4. Download configuration according to the requirements
5. Click download

6. Statement of attention

WCH official website: https://wch-ic.com/

WCH-LINK instructions for use: https://www.wch-ic.com/products/WCH-Link.html